

2023

YEAR 11
YEARLY
EXAMINATION

Mathematics Standard 2

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Write your Student ID below and on page 6
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
80

Section I — 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 20 minutes for this section

Section II — 70 marks (pages 6–21)

- Attempt Questions 11–28
- Allow about 1 hour 40 minutes for this section

STUDENT ID: _____



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Section I

10 marks

Attempt Questions 1–10

Allow about 20 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1** What is the median in the following data set?

2 3 4 8 10

- A. 2
- B. 4
- C. 8
- D. 10

- 2** How much simple interest is earned on an investment of \$3000 for five years at a rate of 1.5% per annum?

- A. \$45
- B. \$175
- C. \$225
- D. \$300

- 3** What is the solution to the following equation?

$$2x + 7 = 5x + 1$$

- A. $x = 1$
- B. $x = 2$
- C. $x = 3$
- D. $x = 4$

- 4 The movie *Math is Awesome* is being screened at the cinemas. The movie starts at 11:45 am and runs for 115 minutes.

When will the movie end?

- A. 1:00 pm
 - B. 1:20 pm
 - C. 1:40 pm
 - D. 2:00 pm
- 5 Kwon measures the length of a string on a ruler in centimetres to one decimal place.

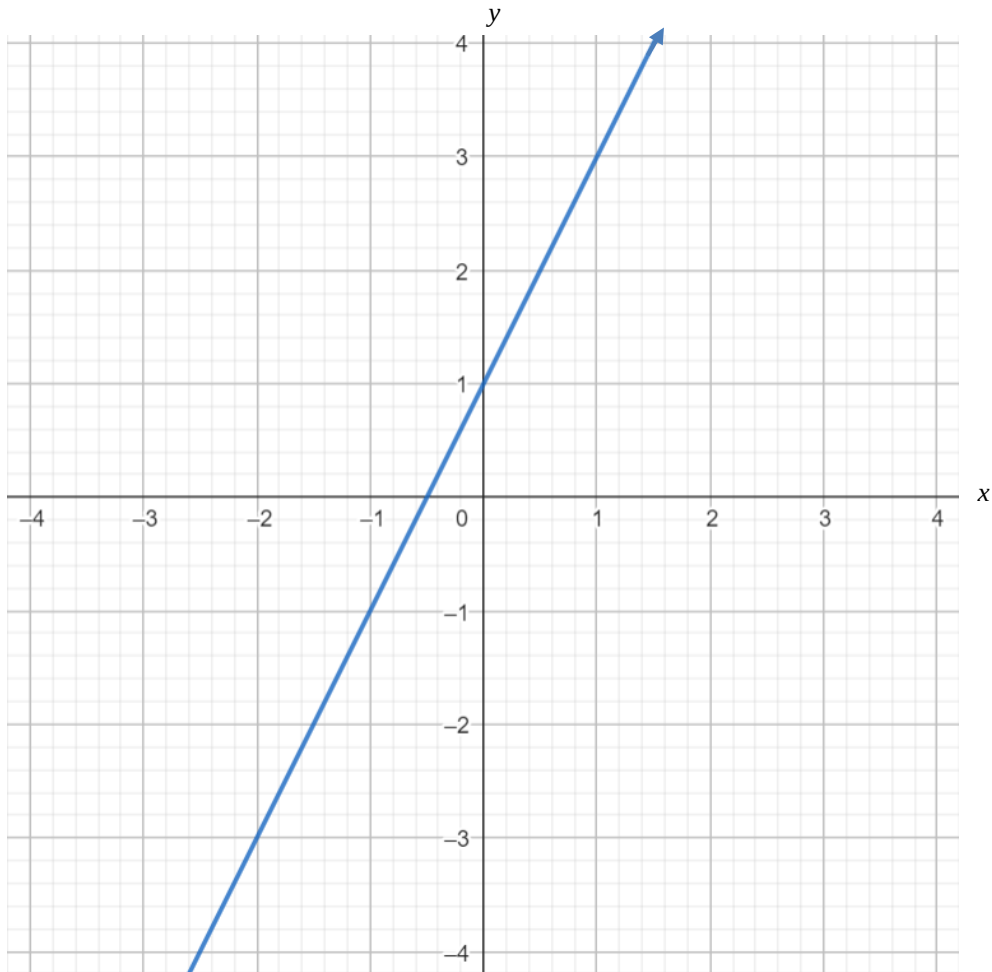


She records the length as 5.3 cm.

What is the percentage error for this measurement?

- A. 0.94 %
 - B. 5.67 %
 - C. 9.4 %
 - D. 10 %
- 6 What is the amount of GST in the price of a television that has a total cost of \$4560?
- A. \$414.55
 - B. \$456.00
 - C. \$506.67
 - D. \$560.00

- 7 The number plane given shows a line.



What is the equation of the line shown on the number plane?

- A. $y = 2x + 1$
- B. $y = \frac{1}{2}x + 2$
- C. $y = -2x + 1$
- D. $y = 2x + 4$

- 8** A jar contains 25 cookies. 18 of these cookies are chocolate chip and 7 are salted caramel.

Lily is to select two cookies from the jar at random.

What is the probability that Lily selects two chocolate chip cookies?

- A. $\frac{18}{25} + \frac{17}{25}$
- B. $\frac{18}{25} \times \frac{17}{25}$
- C. $\frac{18}{25} + \frac{17}{24}$
- D. $\frac{18}{25} \times \frac{17}{24}$

- 9** In a data set, the 5th decile was found to be 75.

Which row shows only true statements about the data set?

A.	60% of the scores are lower than 75.	The 50 th percentile is 75.	The median of the data set is 75.
B.	50% of the scores are higher than 75.	60% of the scores are lower than 75.	The 50 th percentile is 75.
C.	The median of the data set is 75.	50% of the scores are higher than 75.	60% of the scores are lower than 75.
D.	The 50 th percentile is 75.	The median of the data set is 75.	50% of the scores are higher than 75.

- 10** John is driving at 85 km/h and has a reaction distance of 20 m before he applies the brakes.

What is his reaction time?

- A. 0.23 seconds
- B. 0.62 seconds
- C. 0.85 seconds
- D. 1.17 seconds

End of Section I

2023

**YEAR 11 YEARLY
EXAMINATION**

Student ID: _____

Mathematics Standard 2 Section II Answer Booklet

70 marks

Attempt Questions 11–28

Allow about 1 hour 40 minutes for this part

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- Write your Student ID above

Please turn over

Question 11 (2 marks)

If $x = -3$ and $y = 4$, find the value of $2x^2 - \frac{y}{2}$.

2

Question 12 (2 marks)

In the winter months, Amrita uses her 2100 W heater for a total of 23 hours each week.

2

What is the cost of using the heater for a week if electricity is \$0.19 per kWh?

Question 13 (5 marks)

Blood alcohol content (BAC) is calculated using the following formula:

$$BAC_{Male} = \frac{10N - 7.5H}{6.8M}$$

where N is the number of standard drinks consumed, H is the number of hours of drinking, and M is the person's weight in kilograms.

A glass of wine contains 1.2 standard drinks.

Peter weighs 73 kg and consumes three glasses of wine between 2 pm and 4:00 pm.

- (a) Find Peter's BAC to three decimal places.

3

- (b) The time needed for BAC to reach zero after the last drink consumed can be calculated using the following formula:

2

$$\text{Number of hours for BAC to reach zero} = \frac{BAC}{0.015}$$

How long does Peter need to wait until his BAC reaches zero? Answer to the nearest minute.

Question 14 (3 marks)

Omar travels from Sydney (34° S, 151° E) to visit his mother in Morocco (31° N, 7° W). **3**

If he boards his flight on Monday at 3:00 pm, what day and time is it in Morocco?

Question 15 (2 marks)

Calculate the number of kilocalories (kcal) in a packet of peanuts containing 17.6 grams of fat, correct to two decimal places? **2**

Use the following conversions:

1 gram of fat contains 37 kilojoules (kJ) and 1 kilocalorie = 4.184 kilojoules.

Question 16 (2 marks)

A patient received 200 mL of a medicine in 1 hour and 45 minutes through an intravenous drip at a constant rate. His next dose of medicine will be 275 mL. **2**

How long in hours and minutes will it take for the patient to receive the next dose administered at the same rate?

Question 17 (5 marks)

Charlotte plays soccer and has collected data on the number of goals her team scored in the past ten games. She recorded the results in a frequency distribution table, shown.

Goals scored (x)	f	cf	fx
0	2		
1	3		
2	4		
3	2		
4	1		
Total	12		21

(a) Complete the table by filling the cumulative frequency (cf) and (fx) columns. **2**

(b) Show that the mean is 1.75 goals scored. **1**

(c) Find the median number of goals scored. **1**

(d) Would the mean or median be the best measure of average for this data set? Give a reason for your answer. **1**

Question 18 (4 marks)

Trent is a 24-year-old male who plays hockey for 1h 10 min every day using 2687 kJ of energy. The PAL for playing hockey is 6.5.

- (a) Using the formula below, show that Trent's BMR is 8504 kJ/day. 2

$$\{\text{energy out} = (\text{duration of activity}) \times (\text{BMR in kJ/hour}) \times \text{PAL}\}$$

- (b) Using the Schofield formula below and Trent's BMR of 8504 kJ/day, find Trent's weight. 2

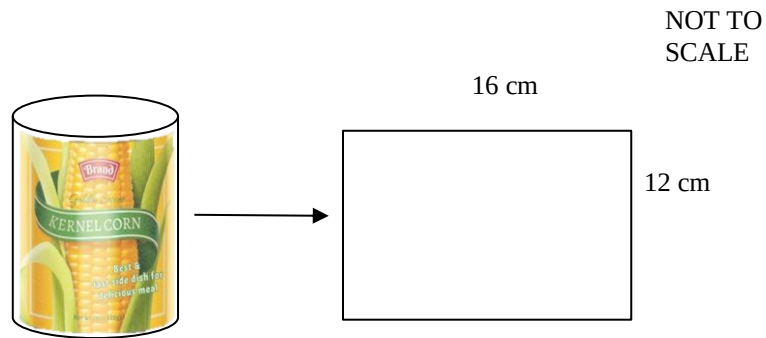
The Schofield Formula for using weight W (kg) to estimate BMR, in kJ/day

Age	BMR (kJ/day)	
	Females	Males
3	$244 \times W - 130$	$249 \times W - 127$
3–10	$85 \times W + 2033$	$95 \times W + 2110$
11–18	$56 \times W + 2898$	$74 \times W + 2754$
19–30	$62 \times W + 2036$	$63 \times W + 2896$
31–60	$34 \times W + 3538$	$48 \times W + 3653$
> 60	$38 \times W + 2755$	$49 \times W + 2459$

Question 19 (5 marks)

A tin of corn kernels has a paper label wrapped completely around the side with no overlap.

The label is removed from the can and laid flat. The dimensions of the rectangular label are shown in the diagram below



- (a) Calculate the radius of the tin correct to two decimal places.

2

- (b) Calculate the total surface area of the tin in terms of π .

3

Question 20 (5 marks)

Shannon earns a salary of \$97 600 and has allowable deductions of \$5178.

The table below shows the tax payable on taxable income.

<i>Taxable income (\$)</i>	<i>Tax payable</i>
\$0 – \$18 200	Nil
\$18 201 – \$45 000	19 cents for each \$1 over \$18 200
\$45 001 – \$120 000	\$5092 plus 32.5 cents for each \$1 over \$45 000
\$120 001 – \$180 000	\$29 467 plus 37 cents for each \$1 over \$120 000
\$180 001 and over	\$51 667 plus 45 cents for each \$1 over \$180 000

- (a) Calculate how much income tax Shannon should pay, not including the Medicare Levy. **3**

- (b) In addition to the tax above, Shannon must also pay the Medicare Levy of \$1848.44. Shannon's employer is withholding \$417.38 per week for tax. **2**

Will she receive a refund at the end of the financial year, or will she have to pay extra tax? Justify your answer with calculations and assume 52 weeks in a year.

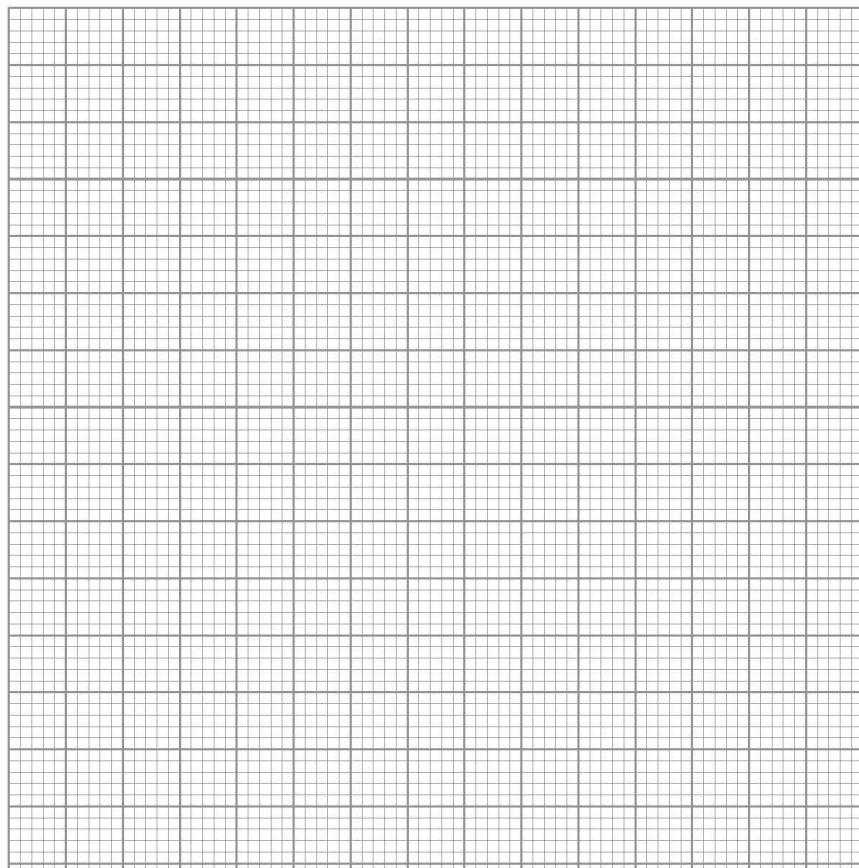
Question 21 (4 marks)

Juan uses a taxi to travel from one town to another. The cost (\$ C) of a taxi ride for k kilometres is given by the formula $C = 6k + 8$.

- (a) How much would it cost to travel 12 km? **1**

- (b) Calculate the maximum distance Juan could travel with \$50. **1**

- (c) Sketch the linear relationship between cost (C) and number of kilometres (k) on the grid below. **2**



Question 22 (7 marks)

Twenty students have a long-term experiment they are working on in their science class.

Data is collected about the number of minutes they spend working on the experiment during each lesson. The set of data collected is displayed in the stem and leaf plot below.

Time spent on experiment in science lesson.

1		3
2		6 6 7 7 9 9 9 9
3		0 2 2 3 4 6 8 8 9
4		2 4

- (a) Calculate the mean time spent working on the experiment. **1**

- (b) Calculate the standard deviation to three significant figures. **1**

- (c) What is the interquartile range of the data? **3**

- (d) Is 13 an outlier for this set of data? Justify your answer using suitable calculations. **2**

Question 23 (3 marks)

Part of a timetable for the Central Coast train line is shown below.

Station	am	am	am	am	pm
Newcastle	8:15	9:09	10:15	10:29	12:10
Hamilton	8:25	9:19	--	10:39	12:30
Broadmeadow	8:33	9:27	--	10:47	12:47
Adamstown	8:45	9:39	--	10:59	12:59
Katora	8:52	9:46	--	11:06	1:06
Cardiff	8:59	9:53	--	11:13	1:13
Cockle Creek	8:04	--	--	11:18	--
Teralba	9:10	10:00	--	11:22	1:20
Booragul	9:15	10:05	11:00	11:28	1:25
Fassifern	9:21	10:11	--	--	1:31
Awaba	9:27	10:18	--	11:37	1:27
Dora Creek	9:33	11:23	--	11:43	12:43
Morriset	9:40	11:30	--	11:50	12:50
Wyee	9:47	10:38	--	11:57	12:57
Warnervale	9:55	10:45	--	--	2:05
Wyong	10:04	10:51	11:25	--	2:08

- (a) Helen wants to catch a train from Cardiff Station to get to Warnervale Station before 11:00 am. 1

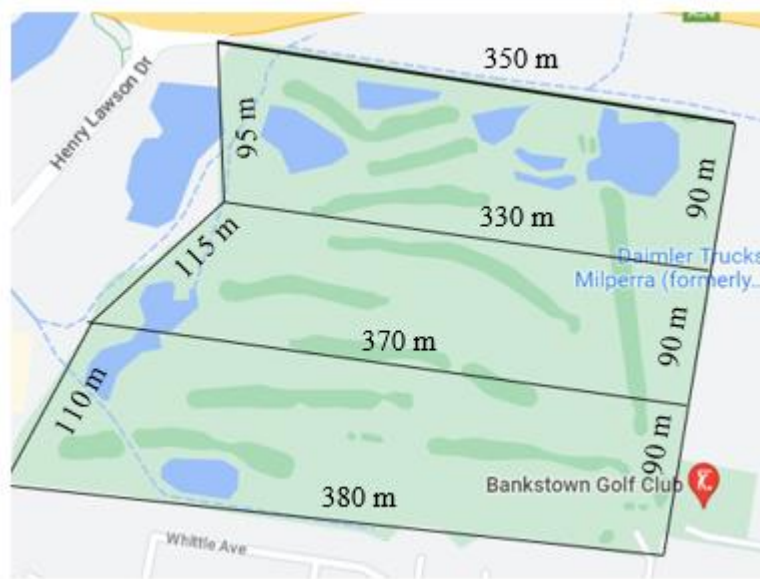
What is the latest time she can catch her train?

- (b) Tammy leaves her friend's house at Cockle Creek to be at the train station at 11 am and wants to get home as quickly as possible. She lives in Awaba and her home is a six-minute walk from the station. 2

Describe which train(s) she could catch to get home and the give the earliest time that she would arrive home.

Question 24 (5 marks)

Pictured below is a map of Bankstown Golf Club. All measurements are in metres.



NOT TO SCALE

- (a) Use three applications of the trapezoidal rule to estimate the area of the golf course. 3

- (b) Occasionally, in heavy rain the golf course floods completely. 2

What is the volume of water, in litres, covering this golf course if the water rises to 3 cm above ground level?

Question 25 (4 marks)

Jim purchases his first vehicle for \$16 800. The dealer charges a delivery fee of \$675 and Service NSW charges a car registration fee of \$480. Jim also takes out comprehensive insurance that costs him \$1260 for the first year.

- (a) Calculate the total cost of purchasing the car. **1**

- (b) Jim's car uses 15 litres of petrol to travel 175 km. Petrol costs 205 cents per litre. **3**

Calculate the maximum distance Jim can drive using \$50 dollars of petrol. Write your answer to the nearest km.

Question 26 (5 marks)

Hailey is a casual employee at a department store and is paid at the following rates:

<i>Weekday Rate</i>	\$26 per hour
<i>Saturday Rate</i>	Time-and-a half
<i>Sunday Rate</i>	Double time

Hailey's timesheet for last week is shown.

Day of the week	Start	Finish	Lunch break (unpaid)
Wednesday	9:30 am	5:00 pm	45 minutes
Saturday	8: 00 am	4:00 pm	60 minutes
Sunday	10:00 am	3:00 pm	20 minutes

- (a) What is Hailey's gross pay for last week?

4

- (b) Superannuation is contributed by her employer into Hailey's trusted fund.

1

How much superannuation is added to her fund if 10.5% of her gross pay is contributed for the week?

Question 27 (4 marks)

A box contains 15 red counters and 25 blue counters. Each counter is numbered with either an even or an odd number. Eight of the red counters and twelve of the blue counters are odd.

- (a) Complete the table below.

2

	<i>Even</i>	<i>Odd</i>	<i>Total</i>
<i>Red</i>			
<i>Blue</i>			
<i>Total</i>			

- (b) One of the counters is chosen at random.

1

What is the probability that the counter is red?

- (c) An odd numbered counter is chosen.

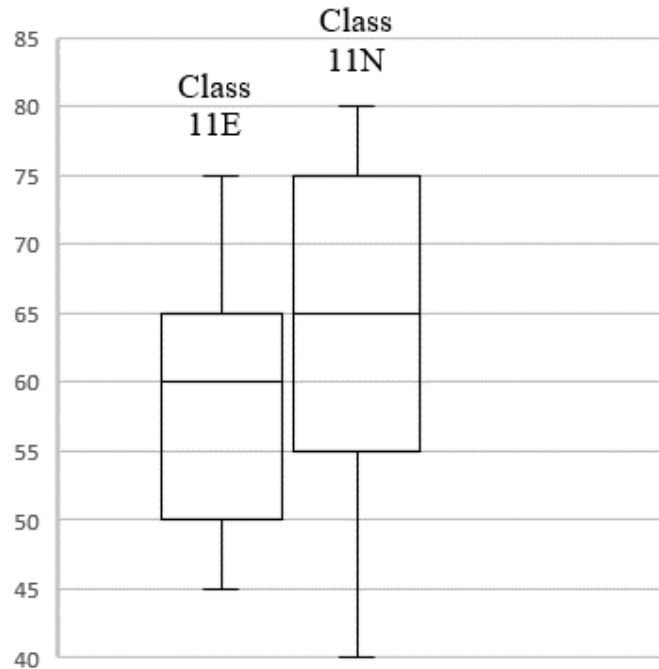
1

What is the probability that the counter is blue?

Question 28 (3 marks)

Two Year 11 classes completed their yearly Mathematics exam. Each class consists of 24 students.

The box and whisker plots show the results of the mathematics exam for each class.



- (a) Describe the shape of the distribution in Class 11E. **1**

- (b) What is the range for class 11E? **1**

- (c) How many students scored below 55 marks in Class 11N? **1**

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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2023 YEAR 11 YEARLY EXAMINATION

Mathematics Standard 1

Mathematics Standard 2

REFERENCE SHEET

Measurement**Limits of accuracy**Absolute error = $\frac{1}{2} \times \text{precision}$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length

$$l = \frac{\theta}{360} 2\pi r$$

Area

$$A = \frac{\theta}{360} \pi r^2$$

$$A = \frac{h}{2} (a + b)$$

$$A \approx \frac{h}{2} (d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3} Ah$$

$$V = \frac{4}{3} \pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \cos A = \frac{\text{adj}}{\text{hyp}}, \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2} ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV (1 + r)^n$$

Straight-line method of depreciation

$$S = V_o - Dn$$

Declining-balance method of depreciation

$$S = V_o (1 - r)^n$$

Statistical Analysis

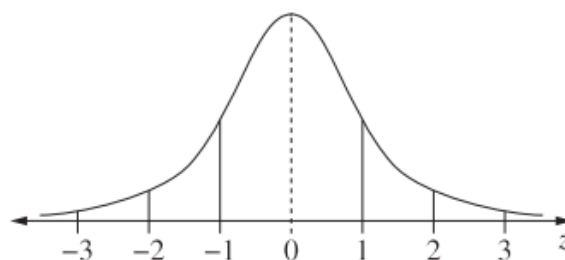
An outlier is a score

less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \bar{x}}{s} \quad z = \frac{x - \mu}{\sigma}$$

Normal distribution

- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of score have z-scores between -2 and 2
- approximately 99.7% of score have z-scores between -3 and 3

2023**YEAR 11 YEARLY
EXAMINATION**

Student ID: _____

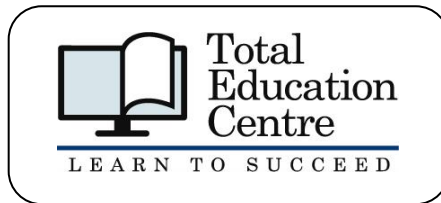
Mathematics Standard 2 Year 11 Examination

Section I Answer Sheet

10 marks**Attempt Questions 1–10****Allow about 20 minutes for this section**

Select the alternative A, B, C or D that best answers the question. Fill in the response circle completely.

- | | | | | |
|-----------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☐ | D ☐ |



2023 Mathematics Standard 2 Year 11 Marking Guidelines

Section I

Multiple-choice Answer Key (Explanations over page)

Question	Answer
1	B
2	C
3	B
4	C
5	A
6	A
7	A
8	D
9	D
10	C

Explanation of Multiple Choice answers

Question	Answer	Explanation
1	B	Median - middle number Arranged in ascending order. $2\ 3\ 4\ 8\ 10$ Therefore, 4 is median.
2	C	Simple Interest = PRN $SI = 3000 \times 1.5\% \times 5$ $SI = \$225$
3	B	$2x + 7 = 5x + 1$ $3x = 6$ $x = 2$
4	C	115 minutes is equivalent to 1 hour and 55 minutes $11:45 + 55 \text{ minutes} = 12:40 \text{ pm}$ $12:40 + 1 \text{ hour} = 1:40 \text{ pm}$
5	A	Precision = 0.1 Absolute Error = $\frac{1}{2} \times 0.1$ = 0.05 Percentage Error = $\frac{0.05}{5.3} \times 100$ = 0.94%
6	A	$\$4560 \div 11 = \414.55
7	A	$y = mx + b$ y-intercept = 1, therefore b = 1 gradient = 2, {sample solution; use coordinates (0,1) and (1,3)} therefore, m = 2 $y = 2x + 1$
8	D	Probability of first selection = $\frac{18}{25}$ Sample space and favourable outcome reduced after first selection. Probability of second selection = $\frac{17}{24}$ Two step experiment: Probabilities are multiplied. Therefore, $\frac{18}{25} \times \frac{17}{24}$

9	D	Decile: cut into 9 values to give 10 equal parts Each decile corresponds to 10% increase to one below it. Option D corresponds to a 6 th decile.
10	C	Convert 85 km/h speed into m/s $\frac{85 \times 1000}{3600} = 23.61 \text{ m/s}$ $d = s \times t$ $20 = 23.61t$ $t = 0.85 \text{ seconds}$

Section II

Note: An incorrect answer in a previous part will not necessarily preclude students from achieving full marks in a later part. Answers here are based on correct prior part answers. Marking will need to adapt to pursue correct method with the use of incorrect prior parts.

Question 11 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Correctly substitutes values into expression	1

Sample answer:

$$2(-3)^2 - \frac{4}{2}$$

$$18 - 2$$

$$= 16$$

Question 12 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Correctly converts watts into kilowatts, or equivalent merit	1

Sample answer:

$$2100 \div 1000 = 2.1$$

$$23 \times 2.1 \times 0.19$$

$$= 9.18$$

It costs \$9.18 to run the heater each week.

Question 13 (5 marks)**a**

Criteria	Marks
• Provides correct calculation	3
• Correctly substitutes into BAC formula	2
• Makes progress to substitute values into BAC formula	1

Sample answer:

$$BAC = \frac{10(3 \times 1.2) - 7.5(2)}{6 \cdot 8(73)}$$

$$BAC = 0.042 \text{ (to three decimal places)}$$

b

Criteria	Marks
• Provides correct calculation	2
• Correctly substitutes BAC value into formula. Allow for error carried forward	1

Sample answer:

$$T = \frac{0.042}{0.015}$$

$$= 2.8 \text{ hours}$$

Therefore, it takes 2 hours and 48 minutes for his BAC to reach zero (to nearest minute)

Question 14 (3 marks)

Criteria	Marks
• Provides correct time and day	3
• Correctly calculates time difference between both cities	2
• Correctly calculates longitude between both cities	1

Sample answer:

$$151 + 7 = 158^0$$

$$158 \div 15 = 10 \text{ hrs } 32 \text{ min difference}$$

Therefore, it is 4:28 am Monday in Morocco.

Question 15 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Correctly converts into kilojoules	1

Sample answer:

$$176 \times 37 = 651.2\text{kJ}$$

$$651.2 \div 4.184$$

$$= 155.64$$

There are 155.64 kcal in the packet to two decimal places.

Question 16 (2 marks)

Criteria	Marks
• Provides correct answer	2
• Makes progress to calculate time, or equivalent merit	1

Sample answer:

$$275 \times 1.75 = 481.25$$

$$\frac{481.25}{200} = 2.4$$

Therefore, it will take 2 hrs and 24 mins for the next dose.

Question 17 (5 marks)**a**

Criteria	Marks
• Provides all correct answers in the table	2
• Provides some correct answers in the table	1

Sample answer:

Goals scored (x)	f	cf	fx
0	2	2	0
1	3	5	3
2	4	9	8
3	2	11	6
4	1	12	4
Total	12		21

b

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\frac{21}{12} = 1.75$$

c

Criteria	Marks
• Provides correct answer	1

Sample answer:

Median = 2

d

Criteria	Marks
• Provides correct answer	1

Sample answer:

Median because data is discrete.

Question 18 (4 marks)**a**

Criteria	Marks
• Provides correct answer	2
• Makes progress substituting values into given formula using conversions	1

Sample answer:

$$1\text{h } 10\text{ min} = \frac{70}{60}$$

$$\left(\frac{70}{60}\right) \times (8504 \div 24) \times 6.5 = 2687 \text{ kJ}$$

Therefore, Trent's BMR is 8504.

b

Criteria	Marks
• Provides correct answer	2
• Correctly substitutes into Schofield formula	1

Sample answer:

$$63 \times W + 2896$$

$$63W + 2896 = 8504$$

$$63W = 5608$$

$$W = 89$$

Therefore, Trent weighs 89 kg.

Question 19 (5 marks)**a**

Criteria	Marks
• Provides correct answer	2
• Provides correct circumference formula and substitutes terms correctly, or equivalent merit	1

Sample answer:

$$C = 2\pi r$$

$$16 = 2\pi r$$

$$r = \frac{16}{2\pi}$$

$$r = 2.5$$

The radius is 2.55 cm to two decimal places.

b

Criteria	Marks
• Provides correct answer	3
• Provides correct SA formula and substitutes terms correctly, or equivalent merit	2
• Attempts to provide SA formula, or equivalent merit	1

Sample answer:

$$SA = 2\pi r^2 + 2\pi r h$$

$$SA = 2\pi(2.55)^2 + 2\pi(2.55)(12)$$

$$= 13.005\pi + 61.2\pi$$

$$= 74.205\pi$$

The surface area of the tin is 74.205π units²

Question 20 (5 marks)**a**

Criteria	Marks
• Provides correct answer	3
• Makes progress to calculate tax payable using tax table	2
• Correctly calculates taxable income	1

Sample answer:

$$\$97600 - \$5178 = 92422$$

$$5092 + (0.325 [92422 - 45\,000])$$

$$= \$20504.15$$

Shannon should pay \$20 505.15 income tax.

b

Criteria	Marks
• Provides correct answer. Allow for error carried forward	2
• Correctly calculates total tax withheld	1

Sample answer:

$$417.38 \times 52 = 21703.76$$

$$21703.76 - (20504.15 + 1848.44) = -648.83$$

Therefore, she owes \$648.83.

Question 21 (4 marks)

a

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$C = 6(12) + 8$$

$$\text{Cost} = \$80$$

b

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$50 = 6k + 8$$

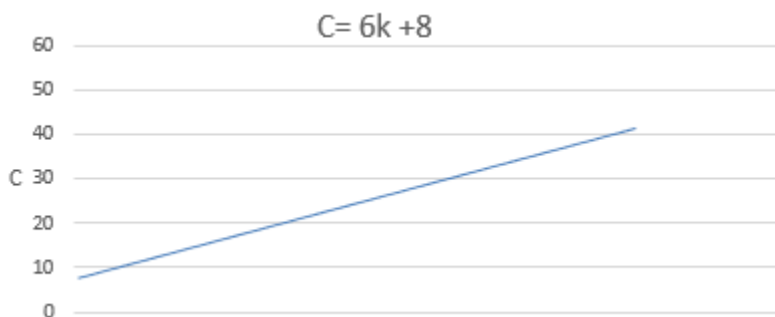
$$k = 7 \text{ km}$$

He can travel a maximum of 7 km.

c

Criteria	Marks
• Provides correct graph	2
• Provides correct value of 8 at vertical axis	1

Sample answer:



Question 22 (7 marks)**a**

Criteria	Marks
Provides correct answer	1

Sample answer:*From calculator*

31.65 is the mean.

b

Criteria	Marks
Provides correct answer	1

Sample answer:*From calculator*

6.73 is the standard deviation to three significant figures.

c

Criteria	Marks
Provides correct answer	3
Makes progress to calculate both upper and lower quartiles	2
Makes progress to calculate upper or lower quartile	1

Sample answer:*Median = 31* *$Q_1 = 28$* *$Q_3 = 37$* *$IQR = 37 - 28$* *$IQR = 9$* **d**

Criteria	Marks
Provides correct answer through justification	2
Makes progress using outlier formula. Allow for error carried forward.	1

Sample answer: *$Q_1 - 1.5 \times IQR$* *$28 - (1.5 \times 9)$* *14.5*

Yes, 13 is an outlier.

Question 23 (3 marks)**a**

Criteria	Marks
• Provides correct answer	1

Sample answer:

9:53 am

b

Criteria	Marks
• Provides correct train and time	2
• Provides correct train or time	1

Sample answer:

Tammy catches the train at 11:18 am. She arrives home at 11:43 am.

Question 24 (5 marks)**a**

Criteria	Marks
• Provides correct answer	3
• Correctly substitutes values into Trapezoidal rule	2
• Makes some progress using Trapezoidal rule	1

Sample answer:

$$A = \frac{1}{2}90(350 + 330) + \frac{1}{2}90(330 + 370) + \frac{1}{2}90(370 + 380)$$

$$A = 30600 + 31500 + 33750$$

$$A = 95\,850 \text{ m}^2$$

The area is approximately 95 850 m².**b**

Criteria	Marks
• Provides correct answer	2
• Makes progress to calculate volume. Allow for error carried forward.	1

Sample answer:

$$95850 \times 0.03$$

$$V = 2875.5 \text{ m}^3$$

$$= 2815.5 \times 1000$$

$$= 2\,875\,500 \text{ L}$$

The volume of water is 2875500 L.

Question 25 (4 marks)**a**

Criteria	Marks
Provides correct answer	1

Sample answer:

$$16800 + 675 + 480 + 1260$$

$$= 19215$$

Total cost of purchasing Jim's car is \$19 215.

b

Criteria	Marks
Provides correct answer	3
Makes progress to calculate distance	2
Correctly calculates litres of petrol for \$50	1

Sample answer:

$$50 \div 2.05 = 24.4L$$

$$(24.4 \times 175) \div 15 = 284$$

The maximum distance Jim's car can travel is 284 km.

Question 26 (5 marks)**a**

Criteria	Marks
Provides correct answer	4
Correctly calculates three days of working hours and rates, or equivalent merit	3
Correctly calculates two days of working hours and rates, or equivalent merit	2
Correctly calculates one day of working hours and rate, or equivalent merit	1

Sample answer:

$$\left(26 \times 6 \frac{3}{4}\right) + (26 \times 1.5 \times 7) + \left(26 \times 2 \times 4 \frac{2}{3}\right)$$

$$175.50 + 273 + 242.67$$

$$\$691.17 \text{ is her gross pay.}$$

b

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$10.5\% \times 691.17$$

\$72.57 is the superannuation contribution.

Question 27 (4 marks)

a

Criteria	Marks
• Provides all correct answers in table	2
• Provides some correct answers in table	1

Sample answer:

	Even	Odd	Total
Red	7	8	15
Blue	13	12	25
Total	20	20	40

b

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\frac{15}{40} = \frac{3}{8} \text{ probability that the counter is red.}$$

c

Criteria	Marks
• Provides correct answer	1

Sample answer

$$\frac{12}{20} = \frac{3}{5} \text{ probability that the odd counter is blue.}$$

Question 28 (3 marks)**a**

Criteria	Marks
• Provides correct answer	1

Sample answer:

The distribution is negatively skewed.

b

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned} \text{Range} &= 75 - 45 \\ &= 30 \end{aligned}$$

c

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\frac{1}{4} \times 24 = 6$$

There are 6 students below 55 marks.

2023 Mathematics Standard 2 Year 11 Yearly Mapping Grid

Section I

Question	Marks	Syllabus outcomes
1	1	S1.2
2	1	F1.1
3	1	MS-A1
4	1	MS-M2
5	1	M1.1
6	1	F1.2
7	1	MS-A2
8	1	MS-S2
9	1	S1.1
10	1	MS-A2

Section II

Question	Marks	Syllabus outcomes
11	2	MS-A1
12	2	M1.3
13	5	MS-A1
14	3	MS-M2
15	2	M1.3
16	2	M1.1
17	5	S1.1
18	4	M1.3
19	5	M1.2
20	5	F1.2
21	4	MS-A2
22	7	S1.2
23	3	MS-M2
24	5	M1.2
25	4	F1.3
26	5	F1.2
27	4	MS-S2
28	3	S1.2